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From search results to generative answers: A mixed-methods study of credibility assessment and trust formation in AI Overviews

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ABSTRACT

Generative search shifts information evaluation from comparing ranked documents to assessing synthesised answers. This study examines how users' perceptions of answer quality are associated with trust and credibility in Google AI Overviews and explores whether visible interface cues may foster trust without systematic source verification. Study 1 analysed survey data from 510 Spanish users using a consistent partial least squares structural equation modelling approach. Study 2 combined eye tracking and interviews with 20 Spanish users. The model explained 75.7% of the variance in trust and 91.0% of the variance in credibility. Clarity and context (β = .419, p < .001) and access to additional information and supporting sources (β = .335, p = .002) showed the strongest observed associations with trust, whereas response immediacy was not significant (β = .008, p = .874). Trust (β = .834) and perceived usefulness (β = .229) were positively associated with credibility (both p < .001), which, in turn, was associated with recommendation following (β = .877) and intention to use (β = .553; both p < .001). Eye tracking recorded no analysable fixations on the AI-error warning, while the interviews indicated that participants in this subsample did not routinely verify the displayed sources and often interpreted visible references as indicators of reliability. Taken together, Study 1 identifies associations between perceived answer-quality cues, trust, and credibility, whereas Study 2 suggests that visual prominence and visible references appeared to operate as authority cues for some participants and may encourage reliance without active source verification.

1. Introduction

Generative search systems such as Google's AI Overviews (AIO) shift information-seeking from comparing ranked documents to evaluating a synthesised answer presented on the results page. This changes both the object of evaluation and the interaction process: source selection and content synthesis are partly delegated to the system, while users increasingly formulate iterative queries and

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assess integrated responses (Tang et al., 2026; Zhou & Li, 2026). This delegation creates a credibility problem. Information science research shows that users judge credibility based on content, source, and interface cues, including surface and verifiable credibility (Liu, 2004; Rieh & Danielson, 2007). In generative systems, visible citations can increase trust even when they are irrelevant, and users often do not inspect the supporting sources (Ding et al., 2025; Nahar et al., 2025; Tang et al., 2026). Credibility can therefore reflect both evaluation of information quality and heuristic responses to interface design.

Most research on trust in AI examines systems that users choose to engage with, including chatbots, conversational assistants, and recommender systems (Bozkurt & Sharma, 2024; Cheng et al., 2022; Hoff & Bashir, 2015). AIO differ because the generative answer is displayed by default within the search results. This raises the question of whether established trust mechanisms carry over to default exposure and how cognitive quality cues interact with authority cues arising from visual prominence, positioning, and visible references. This study examines how cognitive quality cues, functional expectations, and social influence are associated with trust and credibility in AIO, and how these evaluations relate to subsequent behavioural intentions. It also observes attention and verification practices during interaction with the interface. The central issue is not simply whether users adopt AIO, but how they decide whether a synthesised answer is credible enough to guide subsequent behaviour. This study, therefore, adopts an information behaviour perspective: it examines how users process and evaluate information already presented by a deployed system, rather than proposing, comparing, or evaluating the system itself.

1.1. Research objectives

The primary objective of this research is to explain how users form trust and credibility judgements when AIO present synthesised answers by default. This objective is addressed through three specific objectives: (1) to estimate the associations between cognitive quality cues, functional expectations, social influence, trust, credibility, recommendation following, and intention to use; (2) to examine, on an exploratory basis, whether these relationships vary across levels of personal innovativeness; and (3) to identify how visual attention and reported verification practices reveal the cognitive and interface-based cues used to assess AI-generated answers.

The study contributes in three ways. First, Study 1 estimates how self-reported perceptions of answer quality, including clarity and context and access to supporting information, are associated with trust and credibility. Second, Study 2 examines whether visual prominence and visible references function as authority cues during initial exposure to the interface. Third, the cross-study synthesis distinguishes the availability of transparency and verification cues from their actual use.

2. Theoretical framework and hypotheses

2.1. Theoretical framework

Generative search partially delegates content synthesis, source selection, and answer presentation to an AI system. The resulting evaluation process combines judgements about informational quality with authority cues embedded in the interface. To explain the relationships examined, this study draws selectively on UTAUT2 (Venkatesh et al., 2012) and Uses and Gratifications Theory (U&G) (Katz et al., 1973) as supporting resources. This combination is consistent with previous research integrating U&G with technology-adoption perspectives in AI-enabled services and search-engine contexts (Knuth & Masuhr, 2021; McLean & Osei-Frimpong, 2019). UTAUT2 informs the roles of performance expectancy, effort expectancy, and social influence, whereas Uses and Gratifications Theory helps explain the informational value of clarity and context, and access to supporting material (Blut et al., 2022; Ruggiero, 2000). Both perspectives converge in explaining complementary mechanisms through which users reduce uncertainty when evaluating AI-generated responses. U&G theory accounts for the informational value of clarity and context, and adjustable depth, insofar as these dimensions increase the comprehensibility, relevance, and verifiability of the response (Ruggiero, 2000). UTAUT2, in turn, informs the roles of performance expectancy and effort expectancy, indicating whether the system provides a useful response that can be understood and evaluated without excessive cognitive effort (Venkatesh et al., 2012). Taken together, these informational and functional cues constitute evidence of the system' s competence and reliability, thereby reinforcing trust in its outputs. This reading is consistent with research that extends technology adoption models by incorporating trust as a mechanism to overcome uncertainty in interactions with AI-based services (García de Blanes Sebastián et al., 2022). Consequently, facilitating conditions and hedonic motivation are excluded because they are less directly aligned with evaluating an answer already displayed by default. Social influence is associated with trust rather than with intention to use because peer validation may reduce uncertainty before users form an independent assessment of the system's reliability (Krell et al., 2016); (Biloš & Budimir, 2024). Fig. 1 therefore summarises the proposed sequence: functional expectations, social validation, and cognitive gratifications provide complementary cues that reduce uncertainty, support trust formation, and subsequently inform credibility assessment.

2.2. Transparency and trust formation

Trust refers to expectations about the system' s reliability, whereas credibility concerns the user's assessment of the information it provides (Flanagin & Metzger, 2000; Hoff & Bashir, 2015). In this study, transparency is represented by two distinct resources: clarity and context support comprehension, while adjustable depth provides access to additional detail and supporting sources (Lee & Cha, 2025; Tomlinson & Schnackenberg, 2022). Adjustable depth refers to the ability to access further explanations, citations, and external sources. It can support verification, but the mere visibility of references may also serve as an authority cue when users do

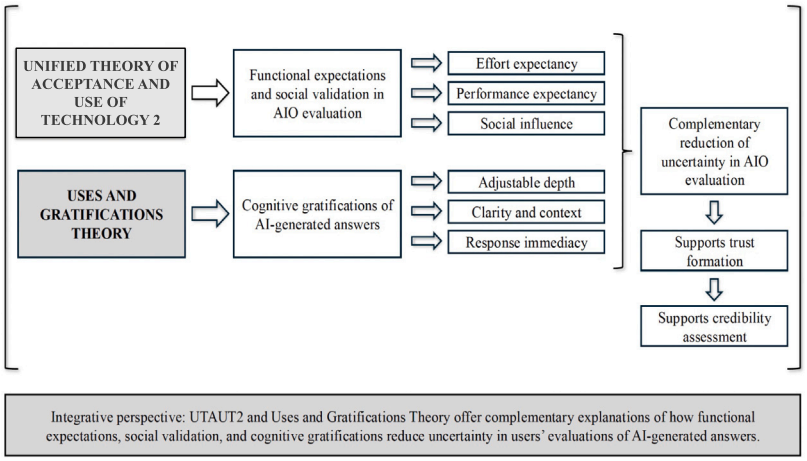


Fig. 1. Integration of supporting theoretical constructs.

not inspect them (Ding et al., 2025; Li & Aral, 2025; Tang et al., 2026). Greater perceived access to supporting information should therefore be associated with stronger trust.

H1: Adjustable depth is positively associated with trust in AIO.

Clarity concerns the comprehensibility of an answer, whereas contextualisation concerns its fit with the user's information need. They are modelled jointly as a reflective construct because both reduce ambiguity and signal answer quality (Lee & Cha, 2025; Schnackenberg & Tomlinson, 2016). Answers perceived as clear and contextually appropriate should therefore be associated with stronger trust.

H2: Clarity and context are positively associated with trust in AIO.

Response immediacy may signal efficiency and technical competence, but its meaning is not uniform: rapid answers may reassure users when information is simple, whereas a slower response can be evaluated more favourably for complex information (J. Kim et al., 2025). The aggregate model, therefore, tests whether a faster perceived response is associated with stronger trust.

H3: Response immediacy is positively associated with trust in AIO.

2.3. Social influence and trust

Social influence can reduce uncertainty by signalling that relevant others consider the technology acceptable and reliable (Krell et al., 2016); (Biloš & Budimir, 2024). In the context of generative search, such social validation may serve as an external cue of credibility, particularly when users have not yet independently evaluated the quality of the generated response. Accordingly, stronger perceived social influence should be associated with greater trust.

H4: Social influence is positively associated with trust in AIO.

2.4. Functional expectations and perceived usefulness

Performance expectancy captures the anticipated improvement in search-task performance, while effort expectancy captures the perceived ease and cognitive efficiency of interaction. Both are expected to be positively associated with perceived usefulness, particularly in natural-language interfaces (Blut et al., 2022; Sugumar & Chandra, 2021; Venkatesh et al., 2003).

H5: Effort expectancy is positively associated with the perceived usefulness of AIO.

H6: Performance expectancy is positively associated with the perceived usefulness of AIO.

2.5. Trust, usefulness, and credibility

Credibility is an attribution concerning the accuracy, reliability, and integrity of information rather than an intrinsic property of the message (Flanagin & Metzger, 2000; Rieh & Danielson, 2007). Trust concerns expectations about the system; credibility concerns the evaluation of its answer. Users who trust AIO should therefore be more likely to regard the information it presents as credible (Hoff & Bashir, 2015; Li & Aral, 2025).

H7: Trust in AIO is positively associated with the perceived credibility of its information.

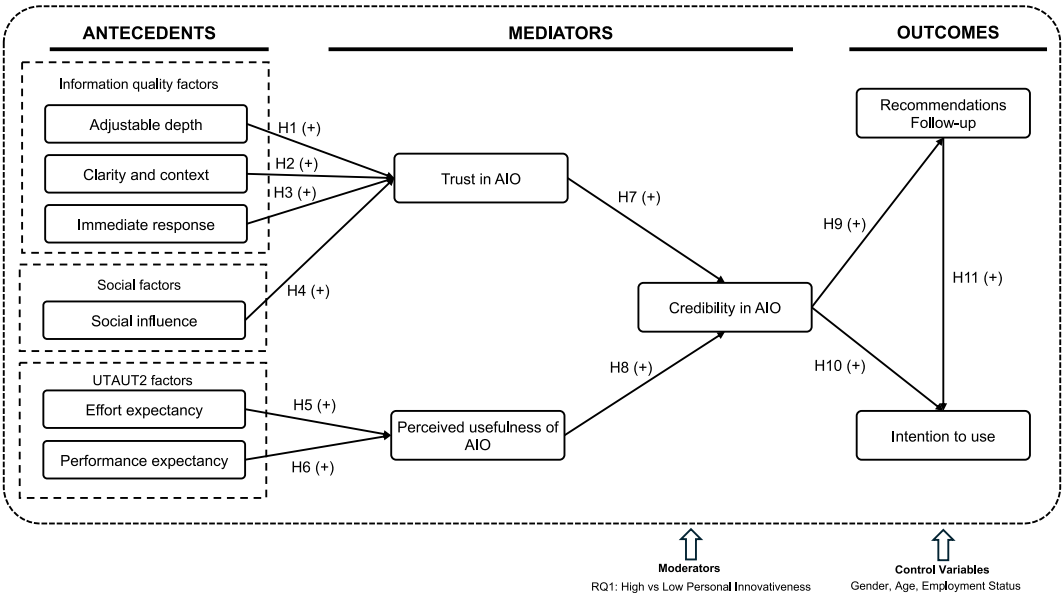


Fig. 2. Conceptual model and hypothesised associations.
Note. Control variables were included in the estimated model but are not visualised as paths.

Perceived usefulness may provide an additional competence cue. When users believe that AIO improves the outcome of a search task, they may also evaluate the information it provides more favourably (Can & Nguyen, 2025; Flavián et al., 2023). Perceived usefulness is therefore expected to be positively associated with credibility.

H8: The perceived usefulness of AIO is positively associated with the perceived credibility of its information.

2.6. Credibility and behavioural intentions

Credibility links information evaluation with behavioural response. Credible information reduces uncertainty and is more likely to be followed, while repeated credible encounters can support future intention to use (Ali et al., 2023; T. Kim et al., 2025; Sussman & Siegal, 2003). Recommendation following may also be associated with future use because acting on an answer creates direct experience with the system.

H9: Perceived credibility is positively associated with recommendation following.

H10: Perceived credibility is positively associated with continued intention to use.

H11: Recommendation following is positively associated with continued intention to use.

2.7. Exploratory analysis of heterogeneity by personal innovativeness

Personal innovativeness in technology refers to an individual's willingness to experiment with new technologies at an early stage of their diffusion (Agarwal & Prasad, 1998; Rogers, 2003). This characteristic may influence how users interpret informational quality, technical efficiency, and social validation when evaluating emerging technologies. However, existing evidence does not establish a uniform direction for these differences across the relationships included in the model. We therefore examine personal innovativeness as a potential source of structural heterogeneity rather than proposing a directional moderating hypothesis (see Fig. 2).

RQ1. Do the structural relationships included in the model differ between users with higher and lower levels of personal innovativeness?

3. Methodology

3.1. Scope of the empirical evaluation

This study does not propose, train, or evaluate an information retrieval system, a natural language processing method, or a large language model. Its object of analysis is user behaviour: specifically, how people form credibility and trust judgements when interacting with a deployed generative search interface. Large language models constitute the technological context of the study, not a method whose performance is measured or optimised. Comparisons with computational baselines are therefore not applicable, because the study advances no claims about algorithmic output quality. This design is consistent with an established

line of information behaviour research published in this journal that examines how users evaluate, select, and act on information produced by search systems without benchmarking those systems against computational alternatives (e.g., Liu (2004), Rieh and Danielson (2007), Tang et al. (2026)). The empirically relevant comparisons in this study are the path coefficients in the structural model, the group differences in the multigroup analysis, and the attentional and verification patterns documented in Study 2.

3.2. Data sources and analytical datasets

Two independently collected datasets were analysed separately and integrated only at the interpretation stage. The Study 1 dataset comprised 510 valid questionnaire records, with the individual respondent as the unit of analysis. It included multi-item construct measures, personal-innovativeness data where available, sociodemographic variables, and indicators of generative AI use. The Study 2 dataset comprised 20 valid participant-level eye-tracking recordings. Each record contained fixation duration, fixation count, and time to first fixation for six predefined areas of interest. The qualitative corpus comprised 60 verbal records collected from the same participants before, during, and after exposure to the interface.

3.3. Study 1 dataset, design, and sample

Study 1 used a cross-sectional online survey and consistent partial least squares structural equation modelling (PLSc-SEM). Data were collected in March 2026 through an online consumer panel managed by ANDATA. Eligible respondents were adult residents of Spain with prior experience using online search engines. All constructs were measured using seven-point Likert scales adapted from validated measures (see Appendix). Before completing the questionnaire, participants received information about the study and its procedure and provided informed consent. The research protocol was approved by the Ethics Committee for Experimentation (CEUMA), Málaga, Spain (approval number CEUMA 47; Registry No. 122-2024-H). Questionnaire records were screened sequentially for eligibility and attention. Respondents first completed a visual check to confirm that they correctly identified an AIO (Li & Aral, 2025). Records with abnormally short completion times were then removed (Benítez et al., 2020), followed by records that failed an instructional manipulation check (Oppenheimer et al., 2009). After screening, 510 questionnaires were retained for the primary structural model. G*Power indicated that 81 observations were required for a medium effect ($f^2 = .15$), 80% power, and $\alpha = .05$; the retained sample therefore provided power above 99% (Hair et al., 2017, 2022). The inverse square root method also indicated sufficient power to detect standardised paths of approximately .10 (Kock & Hadaya, 2018). The final Study 1 sample comprised 510 participants (50.7% women; mean age = 45.41 years, SD = 12.41). Of these, 47.9% were aged 18–44, 27.6% were aged 45–54, and 24.5% were aged 55 or older. Employees represented 70.6% of the sample, and 64.4% had completed higher education.

Most participants reported using GenAI chatbots (90.0%) or GenAI-based translation tools (74.6%). Daily use was reported by 33.7%, and a further 30.1% used these tools two or three times per week. All respondents had previously encountered AIO in search results, and 97.1% reported considering them at least occasionally. These characteristics indicate prior exposure to the focal interface but should not be interpreted as indicative of population representativeness.

3.4. Control of common method bias and assessment of the measurement model

Procedural controls were implemented to mitigate common method bias. Responses were collected anonymously and treated confidentially, while predictor and outcome measures were separated by unrelated questionnaire blocks (Podsakoff et al., 2003, 2024). As a statistical diagnostic, full-collinearity variance inflation factors were below 3 for all constructs, suggesting that common method bias was unlikely to substantially affect the estimated relationships (Koc, 2015). This application of the VIF is distinct from the assessment of predictor collinearity within the structural model reported in Section 3.5. The model was estimated in SmartPLS 4.1 using PLSc, which corrects correlations among reflective constructs to approximate common-factor estimates while retaining a variance-based estimation framework (Hair et al., 2022; Ringle et al., 2024). Statistical inference was based on 10,000 bootstrap subsamples, two-tailed tests, bias-corrected and accelerated (BCa) confidence intervals, and a significance level of $\alpha = .05$.

Following the guidelines of Hair et al. (2022) and Henseler et al. (2015), the analysis was organised in two stages, beginning with the assessment of the measurement model. Individual reliability was verified by confirming that the standardised loadings (λ) exceeded the threshold of 0.708. Internal consistency was examined using Cronbach's alpha (α) and composite reliability (CR), with values above 0.70. Finally, convergent validity was confirmed because the average variance extracted (AVE) exceeded the 0.50 criterion (see Table 1).

Discriminant validity was assessed using the heterotrait-monotrait ratio (HTMT) and the Fornell–Larcker criterion. Table 2 presents HTMT ratios below the main diagonal, latent construct correlations above it, and the square root of each construct's average variance extracted (AVE) in bold along the diagonal. The Fornell–Larcker criterion was satisfied for all construct pairs, as each construct's square root of the AVE exceeded its correlations with the other constructs. Likewise, all HTMT ratios were below the .90 threshold, although values close to this cut-off warrant cautious interpretation. In particular, the HTMT ratio between Performance Expectancy and Perceived Usefulness was .888, while their latent construct correlation was .813, which was lower than the square roots of the AVE for both Performance Expectancy (.878) and Perceived Usefulness (.912). HTMT inference was further examined using 10,000 bootstrap subsamples, a one-tailed test, and bias-corrected confidence intervals. The upper bounds of the resulting confidence intervals, including that for the Trust–Credibility relationship, remained below .90. Overall, these results support discriminant validity, although the construct pairs with HTMT ratios approaching the threshold should be interpreted with caution.

Table 1
Measurement model reliability and convergent validity (n = 510).

Construct	Item	Loading (λ)	α	CR	AVE
Adjustable depth	AD1	0.872	0.920	0.921	0.759
	AD2	0.848			
	AD3	0.909			
	AD4	0.860			
	AD5	0.865			
Clarity and context	CC1	0.838	0.900	0.908	0.769
	CC2	0.882			
	CC3	0.890			
	CC4	0.898			
Credibility in AIO	CR1	0.908	0.917	0.918	0.801
	CR2	0.897			
	CR3	0.887			
	CR4	0.888			
Effort expectancy	EE1	0.902	0.920	0.922	0.807
	EE2	0.895			
	EE3	0.907			
	EE4	0.888			
Response immediacy	RI1	0.827	0.885	0.918	0.813
	RI2	0.937			
	RI3	0.937			
Intention to use	IU1	0.886	0.813	0.814	0.729
	IU2	0.811			
	IU3	0.863			
Perceived usefulness of AIO	PU1	0.908	0.899	0.900	0.833
	PU2	0.925			
	PU3	0.904			
Performance expectancy	PE1	0.840	0.925	0.928	0.770
	PE2	0.867			
	PE3	0.893			
	PE4	0.909			
	PE5	0.877			
Recommendation following	RE1	0.881	0.932	0.932	0.830
	RE2	0.915			
	RE3	0.934			
	RE4	0.914			
Social influence	SI1	0.944	0.945	0.946	0.902
	SI2	0.959			
	SI3	0.945			
Trust in AIO	TR1	0.891	0.921	0.922	0.809
	TR2	0.890			
	TR3	0.916			
	TR4	0.902			

Note. α = Cronbach's alpha; CR = composite reliability; AVE = average variance extracted.

3.5. Structural model estimation results

Once the validity of the measurement model had been established, the structural model was assessed in accordance with the recommendations for PLS-SEM (Hair et al., 2022). Collinearity among the predictor constructs was first examined using variance inflation factors (VIFs), with values below 3.0 taken as evidence that multicollinearity was unlikely to distort the estimated path coefficients. All VIF values were below this threshold (VIF < 3), indicating no problematic collinearity among the structural model predictors. This assessment is distinct from the full-collinearity VIF test reported in Section 3.4, which was used as a diagnostic for potential common method bias.

The model' s explanatory capacity was assessed using R^2, and Stone–Geisser Q^2 values were used to evaluate predictive relevance. Positive Q^2 values indicate that the model provides predictive relevance for an endogenous construct relative to an omission-based benchmark; these values are not out-of-sample PLSpredict results. The hypothesised relationships were evaluated using path coefficients, BCa confidence intervals, and f^2 effect sizes. Effects were interpreted as large ($f^2 > .35$), moderate ($f^2 > .15$), or small ($f^2 > .02$). The model explained 91.0% of the variance in credibility ($R^2 = .910$), 85.4% in perceived usefulness ($R^2 = .854$), 76.9% in recommendation following ($R^2 = .769$), 75.7% in trust ($R^2 = .757$), and 69.5% in intention to use ($R^2 = .695$). The SRMR was .043, below the .08 reference value (Hu & Bentler, 1999). Stone–Geisser Q^2 values were positive for all endogenous constructs, indicating predictive relevance within the estimated model (Table 3).

Table 2
Measurement model reliability and convergent validity (n = 510).

Construct	AD	CC	CR	EE	RI	IU	PU	PE	RE	SI	TR
Adjustable depth (AD)	**.871**	.789	.788	.670	.571	.715	.817	.783	.797	.614	.765
Clarity and context (CC)	.862	**.877**	.811	.749	.664	.682	.787	.747	.755	.526	.754
Credibility in AIO (CR)	.857	.888	**.895**	.661	.550	.707	.766	.751	.811	.604	.874
Effort expectancy (EE)	.727	.830	.718	**.898**	.690	.622	.734	.652	.626	.460	.638
Response immediacy (RI)	.621	.747	.603	.755	**.902**	.567	.647	.582	.492	.290	.535
Intention to use (IU)	.826	.795	.819	.717	.655	**.854**	.779	.737	.686	.555	.748
Perceived usefulness of AIO (PU)	.897	.875	.843	.806	.712	.895	**.912**	.813	.772	.547	.769
Performance expectancy (PE)	.847	.815	.814	.703	.630	.849	.888	**.878**	.750	.552	.756
Recommendation following (RE)	.861	.817	.877	.674	.531	.788	.843	.807	**.911**	.629	.801
Social influence (SI)	.659	.565	.648	.491	.312	.634	.592	.591	.671	**.950**	.613
Trust in AIO (TR)	.830	.823	.897	.692	.583	.864	.845	.817	.864	.656	**.899**

Note. Values below the main diagonal are HTMT ratios; values above the diagonal are latent construct correlations. Bold diagonal values are the square roots of AVE.

To examine whether the high in-sample R^2 values were accompanied by predictive performance beyond the estimation sample, we additionally assessed the model using PLSpredict and the cross-validated predictive ability test (CVPAT). PLSpredict was conducted using 10-fold cross-validation with 10 repetitions and evaluated Q^2predict, RMSE, and MAE relative to the indicator-average and linear-model benchmarks. CVPAT was used to formally compare the model's cross-validated prediction loss against both benchmarks. All endogenous indicators produced positive Q^2predict values, ranging from .348 to .617, indicating that the model's predictions outperformed the mean-based benchmark. For the three indicators measuring Credibility, Q^2predict values ranged from .549 to .578. Their PLS-SEM RMSE values were slightly lower than those obtained from the linear-model benchmark, whereas the MAE differences were small and did not follow a uniform direction.

CVPAT further showed that the prediction loss for Credibility was significantly lower than that of the indicator-average benchmark (difference in loss = −.722, $p < .001$) and did not differ significantly from the linear-model benchmark (difference in loss = −.008, $p = .569$). These results indicate that the predictive performance of Credibility does not collapse outside the estimation sample and is comparable to that of the linear benchmark. Accordingly, the high R^2 should not be interpreted solely as sample overfitting. Nevertheless, these predictive results do not, by themselves, resolve the empirical proximity between Trust and Credibility.

The R^2 of .910 for Credibility is higher than the values reported in related AI-trust research. Zhan et al. (2024) explained 84.1% of trust in healthcare voice AI assistants, Choudhuri et al. (2025) explained 67.9% of trust in GenAI tools, and Tafesse and Sandhu (2026) explained 17.0% of perceived credibility from initial trust. These models are not directly equivalent because their outcomes, predictors, and contexts differ, but the comparison indicates that the present value is unusually high rather than a typical benchmark. In the current model, the primary association reflects the dominant association between Trust and Credibility ($\beta = .834$, $f^2 = 2.211$), with a contribution from perceived usefulness ($\beta = .229$). The PLSpredict and CVPAT results reduce the concern that this value is solely attributable to in-sample overfitting, but they do not eliminate concerns arising from the empirical proximity of Trust and Credibility. Accordingly, $R^2 = .910$ is interpreted as model-specific explanatory power rather than evidence of broadly generalisable predictive accuracy.

Trust was most strongly associated with clarity and context ($\beta = .419$, $p < .001$), followed by adjustable depth ($\beta = .335$, $p = .002$) and social influence ($\beta = .196$, $p < .001$). Response immediacy was not significantly associated with trust ($\beta = .008$, $p = .874$), H3 was therefore not supported. Trust had the largest association with credibility ($\beta = .834$, $p < .001$). Credibility, in turn, was associated with recommendation following ($\beta = .877$, $p < .001$) and intention to use ($\beta = .553$, $p < .001$). Perceived usefulness was also positively associated with credibility ($\beta = .229$, $f^2 = .061$, $t = 5.396$, $p < .001$, 95% BCa CI [0.009, 0.270]); H8 was therefore supported. Performance expectancy ($\beta = .637$, $p < .001$) and effort expectancy ($\beta = .357$, $p < .001$) were positively associated with perceived usefulness. The control checks are reported in Table 3 and are not treated as central structural relationships.

3.6. Multigroup analysis results

An exploratory PLS multigroup analysis (PLS-MGA) examined structural heterogeneity across personal-innovativeness profiles. Personal innovativeness was measured with three items adapted from Agarwal and Prasad (1998) and Rogers (2003). Of the 510

Table 3
Structural model results and hypothesis tests.

Hypothesis	Structural relationship	β	f^2	t	p	95% BCa CI	Decision
H1	Adjustable depth → Trust in AIO	0.335	0.098	3.152	*.002*	[0.141, 0.555]	Supported
H2	Clarity/context → Trust in AIO	0.419	0.140	4.008	< *.001*	[0.207, 0.620]	Supported
H3	Response immediacy → Trust in AIO	0.008	0.000	0.158	.874	[-0.098, 0.111]	Not supported
H4	Social influence → Trust in AIO	0.196	0.086	3.949	< *.001*	[0.097, 0.292]	Supported
H5	Effort expectancy → Perceived usefulness of AIO	0.357	0.441	4.622	< *.001*	[0.208, 0.509]	Supported
H6	Performance expectancy → Perceived usefulness of AIO	0.637	1.405	8.422	< *.001*	[0.486, 0.780]	Supported
H7	Trust in AIO → Credibility in AIO	0.834	2.211	12.392	< *.001*	[0.707, 0.972]	Supported
H8	Perceived usefulness of AIO → Credibility in AIO	0.229	0.061	5.396	< *.001*	[0.009, 0.270]	Supported
H9	Credibility in AIO → Recommendation following	0.877	3.281	36.114	< *.001*	[0.825, 0.920]	Supported
H10	Credibility in AIO → Intention to use	0.553	0.230	5.182	< *.001*	[0.324, 0.742]	Supported
H11	Recommendation following → Intention to use	0.305	0.070	2.817	*.005*	[0.114, 0.534]	Supported
Control relationships	Gender → Recommendation following	0.102		0.224	.823	[−0.121, 0.103]	ns
	Gender → Intention to use	0.102		1.476	.140	[−0.035, 0.237]	ns
	Age → Recommendation following	0.008		0.343	.732	[−0.051, 0.074]	ns
	Age → Intention to use	0.011		0.219	.827	[−0.061, 0.077]	ns
	Employment status→ Recommendation following	−0.003		0.118	.906	[−0.053, 0.046]	ns
	Employment status → Intention to use	−0.051		1.310	.190	[−0.127, 0.026]	ns
	Endogenous construct		R^2		Q^2		
	Trust in AIO		0.757		0.665		
	Perceived usefulness of AIO		0.854		0.725		
	Credibility in AIO		0.910		0.719		
	Recommendation following		0.769		0.646		
	Intention to use		0.695		0.534		

Note. β = standardised path coefficient; BCa CI = bias-corrected and accelerated confidence interval; f^2 = effect size; R^2 = explained variance; Q^2 = Stone–Geisser predictive relevance; ns = not significant. Bootstrap subsamples = 10,000; two-tailed tests.

Study 1 cases, 48 had missing personal-innovativeness data and were excluded from the MGA, leaving 462 cases. Participants scoring above the median were classified as higher-innovativeness users (n = 280), and those scoring at or below the median as lower-innovativeness users (n = 182). Because dichotomising a continuous measure can reduce information and create threshold-dependent differences, the MGA is interpreted as exploratory rather than as a definitive test of moderation. Measurement invariance was assessed using the MICOM procedure, including configural and compositional invariance, as well as equality of composite means and variances (Henseler et al., 2016). The model was then estimated separately for each group. Group differences were examined using non-parametric PLS-MGA, the parametric test, and the Welch–Satterthwaite test, which is robust to unequal variances (Henseler & Fassott, 2010).

The clearest group difference concerned clarity and context ($\Delta\beta$ = −.501, p = .001). This association was stronger among lower-innovativeness users (β = .719, p < .001) than among higher-innovativeness users (β = .219, p < .001), indicating that the two exploratory groups weighted explicit informational cues differently. Response immediacy also differed between groups ($\Delta\beta$ = .202, p = .038, Welch test). It was positively associated with trust among higher-innovativeness users (β = .103, p < .05) but not among lower-innovativeness users (β = −.099, p > .05). This opposing pattern helps explain the non-significant aggregate path,

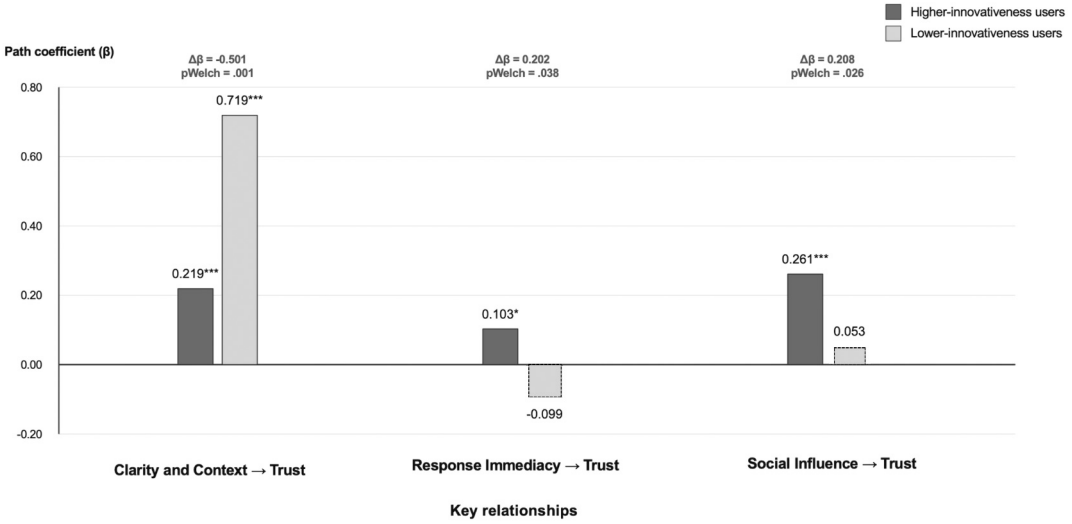


Fig. 3. Path coefficients by exploratory innovativeness profile.
Note. * $p < .05$; *** $p < .001$. Bars with dashed outlines represent non-significant paths.

but should be interpreted with caution given the median split. Social influence showed a further group difference ($\Delta\beta = .208$, $p = .026$, Welch test): the association with trust was significant among higher-innovativeness users ($\beta = .261$, $p < .001$) but not among lower-innovativeness users ($\beta = .053$, $p > .05$). The remaining group comparisons were not statistically significant (Table 4).

A supplemental continuous-moderation analysis was also estimated with 462 observations. Personal innovativeness was modelled continuously as a moderator of the three paths that differed in the exploratory MGA: clarity and context to trust, response immediacy to trust, and social influence to trust. The interaction effects were estimated using the two-stage approach and assessed with 10,000 two-tailed bootstrap subsamples and 95% percentile confidence intervals. The continuous-moderation results were directionally consistent with the MGA. Personal innovativeness weakened the association between clarity and context and trust ($\beta = -.145$, $t = 1.990$, $p = .047$, 95% percentile CI [−.306, −.018], $f^2 = .031$), indicating a stronger association at lower levels of innovativeness. It strengthened the associations between response immediacy and trust ($\beta = .119$, $t = 2.121$, $p = .034$, 95% percentile CI [.020, .240], $f^2 = .031$) and between social influence and trust ($\beta = .110$, $t = 2.416$, $p = .016$, 95% percentile CI [.027, .206], $f^2 = .034$). Because the interaction effect sizes were small, these findings are interpreted as supplemental evidence of directional consistency rather than definitive evidence of moderation (see Fig. 3).

3.7. Discussion of Study 1

Study 1 indicates that trust in AIO is associated primarily with cognitive transparency and informational quality. Clarity and context, together with adjustable depth, were more strongly associated with trust than response immediacy, suggesting that speed alone is not a sufficient cue of credibility (Larsson & Heintz, 2020; Lee & Cha, 2025). Social influence was also associated with trust, consistent with its legitimising role during early adoption (Krell et al., 2016); (Biloš & Budimir, 2024). Trust was strongly associated with credibility, while perceived usefulness showed an additional, smaller positive association. This pattern suggests that credibility judgements depend more strongly on confidence in the system's reliability than on its perceived instrumental value. Nevertheless, the high proportion of variance explained in credibility requires cautious interpretation, as it may partly reflect the conceptual proximity between trust and credibility, even though the PLSpredict and CVPAT results provide evidence that the model retains predictive value beyond the estimation sample.

Credibility, in turn, was associated with recommendation following and intention to use, while recommendation following was also positively associated with continued intention to use. This result suggests that acting on an AIO recommendation may reinforce subsequent usage intentions by translating credibility judgements into direct behavioural experience. The exploratory MGA indicates possible structural heterogeneity. Lower-innovativeness users relied more strongly on clarity and context, whereas higher-innovativeness users appeared more responsive to immediacy and social influence. The supplemental continuous-moderation analysis reproduced these directional patterns, thereby strengthening their internal consistency.

3.8. Study 2 dataset, design, and sample

Study 2 complemented the self-report evidence by observing attention and reported verification during interaction with an AIO interface. It examined which elements attracted attention, whether participants inspected supporting sources, and whether they

Table 4
Exploratory multigroup analysis by personal innovativeness (n = 462).

Structural relationship	Higher PI		Lower PI		Diff.	Group-difference tests		
	β	t	β	t	$\Delta\beta$	M	P	W
Adjustable depth → Trust in AIO	0.381***	5.123	0.155	1.073	0.226	0.167	.210	0.165
Clarity/context → Trust in AIO	0.219***	3.592	0.719***	5.459	−0.501	0.001	.001	0.001
Immediacy → Trust in AIO	0.103*	2.105	−0.099	1.195	0.202	0.041	.084	0.038
Credibility in AIO → Intention to use	0.484***	7.291	0.222	1.296	0.262	0.156	.120	0.155
Credibility in AIO → Recommendation following	0.792***	24.589	0.836***	22.700	−0.044	0.362	.560	0.371
Effort expectancy → Perceived usefulness of AIO	0.299***	3.969	0.435***	6.866	−0.136	0.174	.432	0.167
Performance expectancy → Perceived usefulness of AIO	0.600***	7.750	0.567***	10.068	0.033	0.724	.852	0.729
Perceived usefulness of AIO → Credibility in AIO	0.251***	4.887	0.203*	2.455	0.048	0.619	.696	0.624
Recommendation following → Intention to use	0.283***	3.977	0.450**	2.805	−0.167	0.339	.342	0.340
Social influence → Trust in AIO	0.261***	5.611	0.053	0.657	0.208	0.022	.062	0.026
Trust in AIO → Credibility in AIO	0.668***	13.764	0.754***	10.950	−0.086	0.298	.453	0.309

*** $p < .001$.
** $p < .01$.
* $p < .05$.
Bold values indicate group differences at $p < .05$. $\Delta\beta$ = higher-innovativeness minus lower-innovativeness estimate; M = PLS-MGA; P = parametric group-comparison test; W = Welch–Satterthwaite test.

noticed the warning about possible AI errors. The two datasets were analysed independently and integrated only when interpreting the findings.

Study 2 data were collected in April 2026. Twenty-two adult undergraduate and postgraduate students born between 2000 and 2007 were approached in person near the laboratory at a Spanish university. Eligibility required prior use of online search engines and the ability to explain what an online search engine is. Individuals whose corrective lenses could compromise gaze recording or who could not define an online search engine were ineligible. Two recordings were subsequently excluded because the accuracy error exceeded 1° for either eye, leaving 20 participants (10 women and 10 men). All were familiar with AIO, and 85% reported using them at least once per day. Participation was voluntary and uncompensated. Participants provided written informed consent using a standard form approved by [institutional ethics committee omitted for anonymous review]. Each individual session lasted approximately 10 min and took place in a quiet, isolated, windowless laboratory with controlled lighting and identical equipment and settings. The focus on Generation Z allows examination of users with relatively high exposure to GenAI but does not permit generalisation to other age groups. The final sample size for Study 2 is consistent with the resource-intensive nature of eye-tracking research. In a review of 41 articles applying eye tracking to the study of search engine results pages, Lewandowski and Kammerer (2021) reported a mean sample size of 28 participants. Accordingly, Study 2 was designed to provide explanatory and contextual evidence rather than population-level estimates.

Eye movements were recorded with a Tobii Pro Nano screen-based eye tracker operating at 60 Hz and processed in Tobii Pro Lab 25.23.1545 for Windows. Before the task, each participant completed a nine-point calibration and validation procedure. A recording was retained only when the accuracy error did not exceed 1° for either eye; two recordings failed this criterion. Fixations were identified using the default Tobii I-VT Fixation Filter. Blinks and brief signal losses were treated as missing gaze samples and were not classified as valid fixations. Six mutually exclusive areas of interest (AOIs) were manually delimited on a screenshot before the results were examined: Search bar, Main AI-generated content, Additional information, Related links, Expand answer control, and AI-error warning (Fig. 4). All participants viewed the same simulated search using the Spanish-language query ‘como moverse en roma’ (‘how to get around Rome’) for 30 s on a 24-inch Full HD monitor at 60 Hz. The standardised static display and fixed exposure time ensured that the content, layout, and interface cues remained identical across participants. This controlled presentation allowed initial visual attention across predefined areas of interest to be compared without variation arising from scrolling, clicking, or query reformulation. Participants could visually inspect the answer, references, and interface controls, but they could not expand the answer or open the displayed sources during the task. Browser, zoom, font size, resolution, interface configuration, equipment, and lighting were held constant. Participants verbalised their thoughts during exposure and completed semi-structured interviews before and after the task.

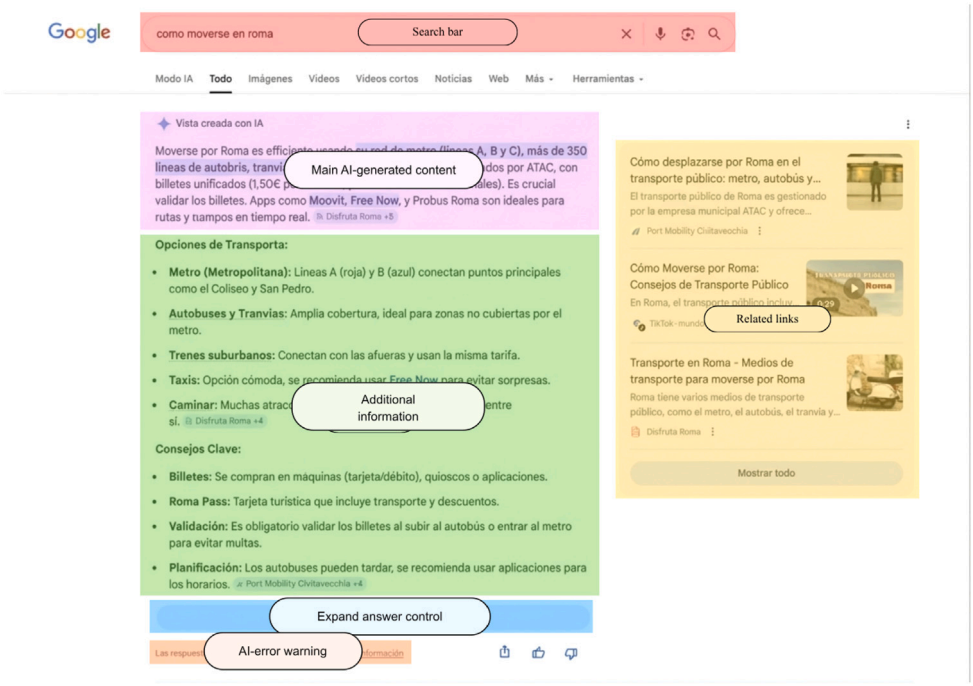


Fig. 4. Areas of interest used in Study 2.

Inferential comparisons focused on five content- and verification-related AOIs: Main AI-generated content, Additional information, Expand answer control, Related links, and the AI-error warning. The Search bar was excluded because it was not part of the generated answer or its verification cues. Total fixation duration and fixation count were compared across the five AOIs using Friedman tests, with Kendall's W as the omnibus effect size. Pairwise post hoc comparisons used two-sided Wilcoxon signed-rank tests, Holm-adjusted p values, and rank-biserial correlations (r_rb). A value of zero was assigned when a participant did not fixate on an AOI. Time to first fixation was compared only for the Main AI-generated content, Additional information, and Related links because the Expand answer control was fixated by only three participants and the AI-error warning by none.

3.9. Eye-tracking results

Three AOI-level metrics were analysed: total fixation duration, fixation count, and time to first fixation. The Main AI-generated content received the longest fixation duration (M = 13.41 s, SD = 5.45), the most fixations (M = 18.10, SD = 9.40), and the shortest time to first fixation (M = .65 s, SD = .78). Additional information ranked second in fixation duration (M = 7.96 s, SD = 4.68) and fixation count (M = 11.25, SD = 10.08), but was first fixated later (M = 8.50 s, SD = 7.34). The Main AI-generated content, therefore, attracted earlier and more sustained attention than Additional information.

Related links received a mean fixation duration of 2.87 s (SD = 2.01), 6.16 fixations (SD = 4.50), and a time to first fixation of 2.61 s (SD = 5.49). The corresponding values for the Search bar were 1.99 s (SD = 1.69), 2.77 fixations (SD = 1.54), and 3.20 s (SD = 3.51). The Expand answer control received the least attention among fixated AOIs: mean fixation duration = .64 s (SD = .60), mean fixation count = 1.33 (SD = .58), and mean time to first fixation = 9.55 s (SD = 6.30). No analysable fixation occurred on the AI-error warning. The aggregate fixation-count heatmap pooled across all participants showed an F-shaped reading pattern concentrated on the first lines of the Main AI-generated content (Fig. 5). The heatmap was based on the total number of fixations recorded across participants. The colour gradient represents fixation intensity, ranging from green for areas with fewer fixations, through yellow for intermediate fixation levels, to red for areas with the highest concentration of fixations.

Friedman tests showed that attention differed across the five focal AOIs for total fixation duration ($\chi^2(4) = 70.95$, $p < .001$, Kendall's W = .887) and fixation count ($\chi^2(4) = 65.27$, $p < .001$, W = .816). Holm-adjusted Wilcoxon comparisons showed that Main AI-generated content had a longer total fixation duration than Additional information (p = .041, r_rb = .590), which in turn exceeded Related links (p < .001, r_rb = .971). Main AI-generated content also received more fixations than Related links (p = .001, r_rb = .924), although the fixation-count differences between Main AI-generated content and Additional information and between Additional information and Related links did not remain significant after correction (both p = .083). The three content-bearing AOIs exceeded the Expand answer control and AI-error warning on both metrics, whereas the latter two areas did not differ significantly. Time to first fixation also differed across the three analysable AOIs among the 19 complete cases ($\chi^2(2) = 10.53$, p = .005, W = .277). Main AI-generated content was fixated earlier than Additional information (p = .002, r_rb = −.895); the other latency comparisons were nonsignificant. Complete pairwise results are reported in Table 7 in the Appendix.



Fig. 5. Aggregate eye-tracking heatmap.
Note. Fig. 5 presents an aggregate fixation-count heatmap pooled across all participants. Colour encodes fixation density, which increases from light green (lowest) through dark green and yellow to red (highest). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3.10. Qualitative analysis of the in-depth interviews

The qualitative corpus comprised 60 verbal records: a pre-exposure interview, concurrent verbalisation, and a post-exposure interview for each participant. All interviews and concurrent verbal reports were conducted in Spanish. Quotations reported in the manuscript were translated into English by the research team. Following an inductive thematic approach (Braun & Clarke, 2006), two researchers independently coded the corpus, compared their coding, refined a shared category structure, and resolved disagreements through discussion until consensus. The same semi-structured protocol was used for all participants and is available from the corresponding author upon reasonable request. Three themes were identified. Theme development was interpretive rather than frequency-based. To make corpus coverage transparent, we report the number of participants and coded excerpts associated with each final theme and identify discrepant or ambiguous cases. A coded excerpt was defined as a substantive participant turn containing evidence relevant to a theme; brief affirmative responses to interviewer probes were not counted as separate excerpts. These counts are descriptive and should not be interpreted as measures of thematic importance. No pre-consensus inter-rater agreement coefficient was calculated. Because agreement calculated after consensus would be uninformative, no retrospective coefficient was added.

Theme 1: Visible references as perceived verifiability cues. Sixteen participants contributed 20 coded excerpts indicating that displayed sources, links, or references increased perceived reliability or certainty. One participant explained, ‘The links showing where the information came from give me a lot of confidence [...] and allow me to go in and check it’ (P6). Another explicitly recognised the heuristic character of this response: ‘Maybe it is a mental bias, but seeing a source or link makes me think that it may be true’ (P15). This pattern was not uniform. P2 stated that no interface element made the answer inherently reliable and reported never using the detail control, whereas P10 identified numerical data rather than references as a reliability cue and said that citations would change his perception very little. Several participants also described active checking through official websites, course materials, or other sources. Visible references are therefore interpreted as cues of perceived verifiability, not as uniform substitutes for verification or evidence that sources were actually opened during the static task.

Theme 2: Default prominence and reduced search effort. Nine participants contributed nine coded excerpts describing the direct, top-positioned answer as convenient or as reducing the need to search further. P13 stated, ‘I no longer have to go from tab to tab [...] it puts the answer there directly.’ P20 described a stronger form of reliance: ‘I do not search for anything else. I trust it, and that is it.’ Discrepant accounts qualified this pattern. P5 initially assumed that the direct answer would be incorrect, P12 reported routinely opening links and seeking greater depth, and P18 found the automatically presented explanation intrusive when seeking a webpage. Default prominence may therefore support convenience-based reliance for some participants, but the interviews do not establish that top position was uniformly interpreted as informational authority.

Theme 3: Limited visibility of the AI-error warning. Seventeen participants contributed 17 excerpts explicitly stating that they had not noticed the warning. P1 stated simply, ‘No, I did not notice it’. Two participants provided clear counter-examples. P2 said,

'Yes, I noticed it, but [...] in the end they are machines and [...] they can fail,' and P5 also reported seeing it. One further account was ambiguous: P7 recalled reading text at the bottom indicating that the answers were generated but said that he had not paid much attention to it, without clearly identifying the error warning. Thus, the interview evidence indicates limited rather than complete invisibility. The two positive reports and the ambiguous case are also retained when interpreting the absence of analysable fixations on the warning AOI, because a lack of classified fixation does not rule out peripheral awareness or later recall.

3.11. Discussion of Study 2

Study 2 provides behavioural and contextual insight into the associations observed in Study 1 by integrating eye-tracking data with interview evidence. The eye-tracking results showed that visual attention was concentrated on the Main AI-generated content, whereas the Additional information and Expand answer control AOIs attracted substantially less attention and the AI-error warning received no recorded fixations. Because the stimulus was static, these findings reflect visual attention to verification-related cues rather than actual source-opening behaviour. Statistical comparisons across AOIs confirmed that the different interface areas attracted markedly unequal levels of visual attention. Specifically, the Main AI-generated content consistently dominated both the duration and timing of participants' fixations.

The interviews complemented these findings by showing that displayed references increased perceived reliability for 16 participants, although reported verification practices included both reliance on visible cues and active checking through external sources. Nine participants described the default direct answer as reducing search effort, although several offered discrepant accounts. Seventeen participants did not report noticing the warning, two reported noticing it, and one described only partial awareness. The discrepancy between these self-reports and the absence of analysable fixations cautions against treating fixation records as evidence of complete non-awareness.

Taken together, these findings distinguish between the availability of transparency cues, the visual attention directed towards them, and their reported use. The combination of increased perceived reliability from visible references, reduced search effort, and limited attention to the warning is consistent with recent research suggesting that users may rely on AI-generated information without systematically examining the evidence supporting it (Li & Aral, 2025; Tang et al., 2026). This pattern indicates a potential risk of reliance without active verification, although it was not uniform across participants. Because the study did not compare participants' trust judgements with the objective accuracy of the displayed information, the findings should not be interpreted as evidence of trust calibration or miscalibration.

4. Discussion of results and implications

The two studies serve different inferential roles. Study 1 examines the proposed structural associations in a broader and more heterogeneous consumer sample, whereas Study 2 provides contextual and process-level evidence from a small, homogeneous group of Generation Z university students with high prior use of generative AI. Their integration should therefore be understood as methodological triangulation and conceptual convergence rather than demographic replication, consistent with mixed-methods eye-tracking research that combines larger quantitative samples with smaller qualitative subsamples to provide contextual and process-level evidence (Anders et al., 2024). The Study 2 participants' age, university status, and familiarity with generative AI may have reduced interface-learning demands and influenced their expectations, attention to credibility cues, and verification strategies. Consequently, convergence between the studies supports the proposed interpretation of the underlying processes but does not imply that identical patterns would occur across age groups or levels of technological experience.

Recent research in information science shows that generative AI is changing information seeking and knowledge construction (Tang et al., 2026; Wei & Chen, 2026). The present findings extend this work by explaining how cognitive evaluations and interface cues coexist when a generative answer is displayed by default. The principal conclusions are drawn on distinct but complementary evidence. The Study 1 structural model supports the associations linking informational evaluations with trust, credibility, recommendation following, and intention to use. Study 2 eye-tracking data demonstrate unequal attention allocation across interface areas, with attention concentrated on the generated answer and little to no attention directed to the Expand answer control and AI-error warning. The interviews show that visible references often increased perceived reliability, while reported verification practices varied and included clear counterexamples to reliance on heuristics. They also indicate that most, but not all, participants failed to report the warning. Together, these sources identify a qualified gap between the availability of verification cues, attention to them, and their actual or reported use. This integration constitutes methodological triangulation at the interpretation stage rather than demographic replication or evidence that both studies measure the same mechanism.

4.1. Theoretical implications

First, this study clarifies how the object of information evaluation changes when users move from comparing ranked documents to assessing a synthesised answer. In generative search, users evaluate a single integrated response that combines content, presentation, and source cues. It distinguishes clarity and context, which support comprehension, from adjustable depth, which provides access to supporting information. These dimensions show that transparency in generative search is multidimensional rather than a single system property (Lee & Cha, 2025; Tomlinson & Schnackenberg, 2022).

Second, the findings distinguish cognitive quality cues from interface-induced authority heuristics. Visual prominence and visible references may support credibility judgements even when users do not inspect the underlying sources. This extends credibility

research by showing that surface-level, reputed, and verifiable cues remain relevant in generative search but are recombined within a single synthesised answer (Liu, 2004; Rieh & Danielson, 2007). Perceived transparency may therefore strengthen trust without necessarily encouraging systematic verification. However, Study 1 did not directly measure the authority heuristic, and the association between adjustable depth and trust should not be interpreted as direct evidence of this mechanism.

Third, Study 2 identifies interface and attentional conditions that may increase the risk of reliance without active verification. The Main AI-generated content received immediate and sustained attention, whereas the Expand answer control received limited attention and the AI-error warning received none. In terms of the Trustworthiness Assessment Model, interface cues were highly available, whereas cues concerning system limitations were neither detected nor used (Schlicker et al., 2025). This interpretation concerns a potential risk rather than measured miscalibration, because the study did not compare participants' trust with answer accuracy.

Finally, UTAUT2 and Uses and Gratifications Theory provide complementary explanatory lenses rather than constituting the study's primary contribution. UTAUT2 captures functional expectations and social validation, whereas Uses and Gratifications Theory captures the informational gratifications sought from AI-generated answers. Their integration explains how functional, social, and cognitive cues jointly reduce uncertainty and shape trust and credibility assessments.

4.2. Practical implications for generative search design and information practice

Generative-search interfaces should make evidence easier to evaluate rather than merely more visible. Citations should be linked to the specific claims they support, source identity and provenance should be clear, and access to additional detail should not require users to infer whether the evidence is relevant. These measures address the finding that visible references can act as authority cues without prompting verification. Warnings about possible AI errors should have greater visual prominence and appear close to the generated claims to which they apply. The absence of fixations on the current warning indicates that making supporting information available is insufficient to ensure that users notice or use it when assessing an answer. Interfaces could also use progressive disclosure to present concise answers while preserving direct access to evidence, uncertainty, and alternative sources. Information-literacy guidance should encourage users to distinguish a well-structured answer from a well-supported answer. The exploratory heterogeneity findings further indicate that design recommendations should be tested across different user profiles rather than assuming uniform cue weighting.

5. Limitations and future research directions

This study has some limitations regarding its generalisability. Study 1 is cross-sectional, so its paths represent associations rather than temporal or causal effects. The Spanish online panel sample had substantial prior exposure to GenAI and may not represent populations with less experience. The median split used for the exploratory MGA also discards information and can create threshold-dependent group differences. Relatedly, the exceptionally high proportion of variance explained in credibility ($R^2 = .910$) likewise constitutes a limitation that must be acknowledged. An R^2 of this magnitude does not necessarily imply overfitting, although it could stem from a degree of conceptual proximity between credibility and some of its predictors. Accordingly, this result should be interpreted as indicative of substantial explanatory power within the current sample, rather than as evidence of predictive capacity extrapolatable to other contexts.

Study 2 involved a small, demographically homogeneous group of Generation Z university students with extensive prior experience using generative AI, along with a simulated interface and a single 30-second search task. Consequently, the eye-tracking and interview findings should be regarded as explanatory and contextual rather than as population-level estimates, and they should not be directly generalised to older or less experienced users. The Study 2 task involved a lower-stakes travel-planning scenario. Although inaccurate transportation guidance can have practical consequences, the task did not reproduce the perceived cost of error associated with medical, financial, or other high-stakes decisions. The observed attention and verification patterns should therefore not be generalised to contexts in which errors carry substantially greater consequences. Future research should systematically compare low-, moderate-, and high-stakes search tasks. The standardised 30-second static display enabled the comparison of initial visual-attention allocation under identical conditions but did not allow participants to scroll, open citations, expand the answer, or reformulate the query. The findings therefore do not capture subsequent interactive verification behaviour or the complete search process. Future research should examine these behaviours in self-paced, fully interactive search environments. Moreover, the cross-sectional design precludes causal or temporal inferences regarding the relationships among credibility, recommendation following, and intention to use. Although the proposed model specifies theoretically grounded directional relationships, alternative explanations remain plausible. In particular, pre-existing intentions to use the system may have influenced participants' retrospective credibility judgements, especially given the strong correlation between these constructs.

Future research should use longitudinal and experimental designs, more diverse samples and search tasks, continuous interaction tests for personal innovativeness, and direct manipulations of citation relevance, warning prominence, and source-verification controls. Replication across countries and interface versions would also show whether the observed cue-weighting patterns remain stable.

CRediT authorship contribution statement

Yolanda Casermeiro-Corpas: Writing – review & editing, Writing – original draft, Software, Formal analysis, Conceptualization. **Francisco Rejón-Guardia:** Writing – review & editing, Writing – original draft, Visualization, Software, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **María Vallespín:** Writing – review & editing, Writing – original draft, Supervision, Data curation. **Sebastián Molinillo:** Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Funding acquisition, Formal analysis.

Ethical statement

The study protocol was reviewed and approved by the Ethics Committee for Experimentation (CEUMA), Málaga, Spain (approval number CEUMA 47; Registry No. 122-2024-H).

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used Grammarly and OpenAI Codex to edit the language and check for textual consistency. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ipm.2026.105211.

Data availability

Data will be made available on request.

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